

Offline language models for field service - Tessellate AI white paper, 2026

Field technicians often work in basements, plant rooms and lift shafts with no connectivity. Project Wren brings a question-answering assistant to their tablets that runs entirely on the device.

We compared int8 and int4 weight quantization. The int4 model halved memory use again, but its error rate on troubleshooting answers rose from 6.1 percent to 11.7 percent, so Wren ships with int8 weights.

Field trial

A field trial with 120 technicians at a German elevator maintenance company ran for eight weeks. The average time to find the right repair procedure dropped from 9 minutes to 3.5 minutes, and technicians used the assistant on 71 percent of their jobs.

Technicians asked most often for wiring diagrams and error code explanations. Photo-based search is the most requested feature for the next release.